Classical Field Theory — Exercise Sheet 23

A second KdV bracket and further conserved quantities

Hints and answers

Use Lecture 23's field space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$, gradient space $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$ and admissible smooth functionals $\mathcal{A}_{\mathrm{KdV}}$. Write $D_x = \partial_x$ for spatial differentiation. Every time-dependent conclusion is conditional on a C^1 path in $\mathcal{S}$ satisfying KdV.

Exercise 23.1 (A variable Hamiltonian operator). For $u \in \mathcal{S}$ define the linear operator $J_2[u] : \mathcal{B} \to \mathcal{S}$ by

$$J_2[u]b = -b''' + 4ub' + 2u'b.$$

The prime means D_x, including when applied to a gradient. Define $\{F, G\}_2 = \int_{\mathbb{R}} F_u J_2[u] G_u \, \mathrm{d}x$.

(a) Prove skewness by integration by parts with both gradients in $\mathcal{B}$, keeping the endpoint primitive. Derive the gradient of $\{F, G\}_2$ and prove closure of the stated class.

(b) Prove Jacobi, also for $J_2 + \alpha D_x$ with every constant $\alpha \in \mathbb{R}$. Use the Hessian symmetry from Lecture 23 and account for the extra terms from differentiating the operator itself. A useful notation is $W(a, b) = ab' - a'b$.

(c) For $P = \frac{1}{2}\int u^2$ and $H = \int (u^3 + u_x^2/2)$, prove $J_2 P_u = D_x H_u$. Is the Gardner mass Casimir also a Casimir of this bracket?

The second KdV operator is discussed in Olver, *Applications of Lie Groups to Differential Equations*, second edition, Examples 7.6 and 7.9, pp. 438–439 and 444. The operator above states the course's scaling and signs explicitly; its proof is part of this problem.

Hint. For the operator derivative use $D(J_2[\cdot]b)(u)h = 4hb' + 2h'b$ when b is held fixed. After the Hessian terms cancel, show that the cyclic third-derivative terms form the derivative of $W(a, b)c'' + W(b, c)a'' + W(c, a)b''$.

Answer. Writing $a = F_u$, $b = G_u$ and $H_F = D(F_u)(u)$, the bracket gradient is

$$(\{F, G\}_2)_u = H_F J_2 b - H_G J_2 a + 2(ab' - a'b).$$

The bracket is Poisson on the stated class. Both representations give $6uu' - u'''$. The mass is not a J_2 Casimir because $J_2 1 = 2u'$.

Exercise 23.2 (Finding a higher polynomial invariant). For real constants A, B, C put

$$I_{A,B,C}[u] = \int_{\mathbb{R}} \left(Au^4 + Bu(u')^2 + C(u'')^2 \right) \mathrm{d}x.$$

Determine all triples such that $DI_{A,B,C}(u)(6uu' - u''') = 0$ for every $u \in \mathcal{S}$. Conclude conservation along every stated Schwartz KdV path. Show necessity as well as sufficiency; do not infer independence of two integrals merely by looking at different integrands.

Hint. Compute the Euler derivative first. Reduce the resulting derivative to integrals of $u(u')^3$ and $u'(u'')^2$. To separate them use amplitude rescaling and the asymmetric Schwartz profile $(1 + x)e^{-x^2}$.

Answer. All triples are $(A, B, C) = \zeta_4(5, 10, 1)$, with $\zeta_4 \in \mathbb{R}$. In particular, $\int [5u^4 + 10u(u')^2 + (u'')^2] \mathrm{d}x$ is conserved.

Exercise 23.3 (Classifying Gardner Casimirs). A Casimir of Lecture 23's Gardner bracket is an admissible functional commuting with every admissible functional. Prove that these are exactly

$$F[u] = \psi(M[u]), \qquad M[u] = \int_{\mathbb{R}} u \, \mathrm{d}x,$$

for smooth $\psi : \mathbb{R} \to \mathbb{R}$. Constant scalar functionals are included. Explain why this does not say that every local polynomial density is a nonlinear function of the density u. Compare the local-functional discussion in Olver, Example 7.13, p. 446: the present admissible class also allows nonlocal composites $\psi(M)$.

Hint. Bracket with every smeared field $u(g)$, $g \in \mathcal{S}$. Then move the spatial derivative onto the gradient. For the converse and global classification, fix one $\eta \in \mathcal{S}$ with integral one and use the affine mass fibres.

Answer. A Casimir has a gradient constant in x at every field. It is constant on each mass fibre, so $\psi(s) = F[s\eta]$ gives $F = \psi \circ M$. Conversely its gradient is $\psi'(M)$ in $\mathcal{B}$, killed by D_x. This is a classification of functionals, not a pointwise density statement.